

Assignment 9 : Regression and Classification

This assignment is aimed at:

1. Learning to design effective prompts for LLMs to generate full machine learning pipelines.
 2. Importance of structured prompting (steps, metrics, outputs, constraints).
 3. Comparison of multiple classification algorithms under consistent evaluation conditions.
 4. Learning how different models behave on increasingly complex cluster based datasets.
 5. Importance of:
 - Train/test separation consistency.
 - Unified metric storage for fair comparison.
 - Visual decision boundary analysis.
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Tasks:

A) Create a prompt that can be used with any LLM to generate the same results as those obtained in the provided Python notebooks:

- “Assignment_8_multiple_regression_methods.ipynb”

The prompt should guide the LLM to reproduce almost similar outputs.

1. Which LLM did you use?
2. Provide the prompt that you given to LLM.

B) In this task, you will be processing the following datasets:

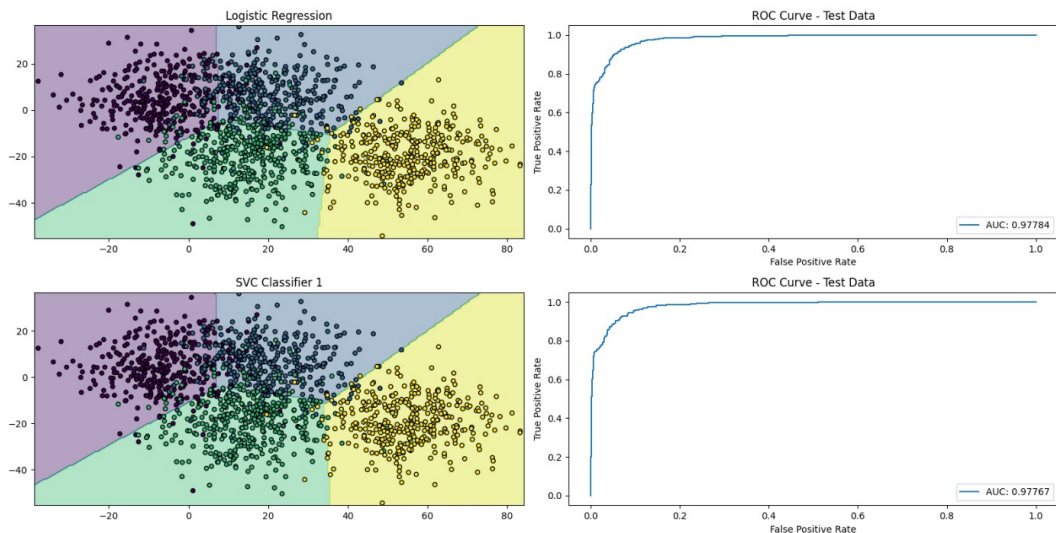
- Clusters-4.csv
- Clusters-5.csv
- Clusters-6.csv

1. Divide each data set into ‘train’ and ‘test’ datasets, once, and use them for all subsequent steps.
2. Use the following algorithms / variants to process the datasets:
 - a. Logistic Regression
 - b. Logistic Regression after adding polynomial features
 - c. SVC - with ‘linear’ kernel (what is ‘linear’?)
 - d. SVC - with ‘rbf’ kernel (what is ‘rbf’?)

- e. Random Forest Classifier - for various combinations of “minimum observations per leaf” (say 1 to 5) and “tree depth” (say 2 to 5). Think of this as a “grid search” problem, where a model is created for every combination of these parameters.
 - f. K-Nearest Neighbours - Try different values of k (e.g., 1, 3, 5, 7, 9) and observe how it affects overfitting vs smoothing.
 - g. Neural Network Classifier - with hidden_layer_sizes=(4)
 - h. Neural Network Classifier - with hidden_layer_sizes=(4,4)
 - i. Neural Network Classifier - with hidden_layer_sizes=(4,4,4)
 - j. Neural Network Classifier - with hidden_layer_sizes=(8)
3. In each of the above cases, generate, capture, and save all the metrics for all the datasets into a common CSV file to facilitate analysis later on. The following metrics (for train and test data) should be created: (For example, see the image that follows):
 - a. Accuracy, Precision (per class), Precision (average), Recall (per class), Recall (average), F1-score (per class), F1-score (average), AUC (per class), AUC (average)
 - b. (Hint: The following functions may be used: accuracy_score, precision_score, recall_score, f1_score, roc_auc_score, roc_curve)

algorithm_name	train_or_test_data	accuracy	precision_1	precision_2	precision_3	precision_4	precision_avg	recall_1	recall_2	recall_3	recall_4	recall_avg	F1_1	F1_2	F1_3	F1_4	F1_avg	AUC_1	AUC_2	AUC_3	AUC_4	AUC_avg
Logistic Regression	train	0.9514	0.9521	0.9470	0.9373	0.9690	0.9513	0.9521	0.9404	0.9406	0.9723	0.9513	0.9521	0.9437	0.9389	0.9706	0.9513	0.9960	0.9955	0.9923	0.9972	0.9952
Logistic Regression	test	0.9549	0.9710	0.9722	0.9333	0.9444	0.9553	0.9853	0.9333	0.9459	0.9577	0.9556	0.9781	0.9524	0.9396	0.9510	0.9553	0.9996	0.9976	0.9978	0.9984	0.9983
SVC Classifier 1	train	0.9540	0.9556	0.9505	0.9406	0.9690	0.9539	0.9589	0.9439	0.9406	0.9723	0.9539	0.9573	0.9472	0.9406	0.9706	0.9539	0.9956	0.9954	0.9924	0.9973	0.9952
SVC Classifier 1	test	0.9549	0.9710	0.9722	0.9333	0.9444	0.9553	0.9853	0.9333	0.9459	0.9577	0.9556	0.9781	0.9524	0.9396	0.9510	0.9553	0.9995	0.9976	0.9974	0.9983	0.9982
SVC Classifier 2	train	0.9497	0.9583	0.9443	0.9247	0.9719	0.9498	0.9452	0.9509	0.9441	0.9585	0.9497	0.9517	0.9476	0.9343	0.9652	0.9497	0.9967	0.9955	0.9920	0.9970	0.9953
SVC Classifier 2	test	0.9583	0.9571	0.9595	0.9351	0.9851	0.9592	0.9853	0.9467	0.9730	0.9296	0.9586	0.9710	0.9530	0.9536	0.9565	0.9585	0.9995	0.9969	0.9975	0.9982	0.9981
Random Forest Classifier 1	train	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Random Forest Classifier 1	test	0.9410	0.9054	0.9571	0.9583	0.9444	0.9413	0.9853	0.8933	0.9324	0.9577	0.9422	0.9437	0.9241	0.9452	0.9510	0.9410	0.9989	0.9903	0.9939	0.9982	0.9953
Random Forest Classifier 2	train	0.9627	0.9723	0.9514	0.9512	0.9757	0.9627	0.9623	0.9614	0.9545	0.9723	0.9626	0.9673	0.9564	0.9529	0.9740	0.9626	0.9993	0.9987	0.9983	0.9996	0.9990
Random Forest Classifier 2	test	0.9479	0.9437	0.9577	0.9467	0.9437	0.9479	0.9853	0.9067	0.9595	0.9437	0.9488	0.9640	0.9315	0.9530	0.9437	0.9481	0.9989	0.9926	0.9969	0.9984	0.9967
Random Forest Classifier 3	train	0.9583	0.9655	0.9507	0.9446	0.9723	0.9583	0.9589	0.9474	0.9545	0.9723	0.9583	0.9622	0.9490	0.9496	0.9723	0.9583	0.9988	0.9980	0.9972	0.9991	0.9983
Random Forest Classifier 3	test	0.9583	0.9710	0.9595	0.9467	0.9571	0.9586	0.9853	0.9467	0.9595	0.9437	0.9588	0.9781	0.9530	0.9530	0.9504	0.9586	0.9989	0.9936	0.9972	0.9985	0.9970
Neural Network Classifier 1	train	0.9071	0.9422	0.9485	0.9508	0.8129	0.9136	0.9486	0.9053	0.8112	0.9619	0.9068	0.9454	0.9264	0.8755	0.8811	0.9071	0.9937	0.9915	0.9710	0.9802	0.9841
Neural Network Classifier 1	test	0.9028	0.9577	0.9710	0.9375	0.7738	0.9100	1.0000	0.8933	0.8108	0.9155	0.9049	0.9784	0.9306	0.8696	0.8387	0.9043	0.9999	0.9942	0.9829	0.9736	0.9876
Neural Network Classifier 2	train	0.9219	0.9507	0.8856	0.9598	0.9010	0.9243	0.9247	0.9509	0.8357	0.9758	0.9217	0.9375	0.9171	0.8935	0.9369	0.9212	0.9878	0.9833	0.9693	0.9896	0.9825
Neural Network Classifier 2	test	0.9236	0.9714	0.9452	0.9538	0.8375	0.9270	1.0000	0.9200	0.8378	0.9437	0.9254	0.9855	0.9324	0.8921	0.8874	0.9244	1.0000	0.9965	0.9873	0.9833	0.9918
Neural Network Classifier 3	train	0.9453	0.9589	0.9462	0.9301	0.9458	0.9452	0.9589	0.9263	0.9301	0.9654	0.9452	0.9589	0.9362	0.9301	0.9555	0.9452	0.9932	0.9916	0.9886	0.9962	0.9924
Neural Network Classifier 3	test	0.9514	0.9706	0.9722	0.9221	0.9437	0.9521	0.9706	0.9333	0.9595	0.9437	0.9518	0.9706	0.9524	0.9404	0.9437	0.9518	0.9994	0.9971	0.9970	0.9969	0.9976

4. For each dataset, generate plots like the ones below - to understand the classification boundaries and the overall performance of the classifier:



5. Compare the metrics within and across the datasets (train and test) and algorithms. In addition to the variations in metrics, compare aspects related to classification boundaries, overfitting, etc.

Submission guidelines:

You are required to submit a single ZIP file named **“yourname_Assignment_9.zip”**. This ZIP file must include the following items:

Task A:

- A PDF file containing the prompt used in the LLM
 - (i) Filename: **“yourname_taskA_assignment_9.pdf”**
- A Jupyter Notebook file
 - (ii) Filename: **“yourname_taskA_assignment_9.ipynb”**

Task B:

- A PDF file containing the analysis of results
 - (i) Filename: **“yourname_taskB_assignment_9.pdf”**
- A Jupyter Notebook file
 - (ii) Filename: **“yourname_taskB_assignment_9.ipynb”**
